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Mini Project – Final Review

Trash to Tech: An AI Model for Intelligent Waste Classification

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Improper waste segregation is a major environmental problem that leads to pollution, health risks, and inefficient recycling, as most existing waste management systems rely on manual sorting or simple rule-based approaches that often result in mixed waste. To address this issue, this project proposes an AI-based waste classification system that automatically categorizes waste images into biodegradable, recyclable, and hazardous classes using a Convolutional Neural Network with transfer learning. The system uses an EfficientNet-based CNN architecture to extract visual features from uploaded waste images, after which it predicts the waste category, generates a confidence score, and provides an appropriate disposal suggestion to guide the user. By automating waste identification and offering disposal guidance, the proposed system reduces human effort, improves segregation accuracy, and can be further extended in the future for integration with smart waste bins and automated waste management systems.

- Waste management is an important challenge in modern cities due to increasing population and consumption.
- Improper segregation of waste leads to environmental pollution and reduces the efficiency of recycling processes.
- Artificial Intelligence and Deep Learning can analyze images and automatically identify waste types.
- This project proposes an **AI-based waste classification system using CNN and transfer learning** to classify waste images and recommend proper disposal methods.

Year	Author(s)	Title	Journal/Conf	Research Design	Conceptual / Theoretical Framework	Proposed Framework	Major Theme	Future Ideas
2021	Chu et al.	Towards artificially intelligent recycling: Improving image processing for waste classification	arXiv	Transfer learning + data augmentation	Transfer learning	Augmentation-enhanced CNN	Boosting accuracy with TL + augmentations	Explore more augmentation strategies, domain adaptation
2021	Dutta et al.	Improving the Deployment of Recycling Classification through Efficient Hyper-Parameter Analysis	arXiv	Lightweight CNN + hyperparameter tuning	CNN deployment strategies	Optimized lightweight CNN	Deployment on embedded/edge devices	Expand to IoT and mobile hardware integration
2021	Zhang et al.	Waste image classification based on transfer learning and convolutional neural network	ScienceDirect	Collected dataset + TL-CNN	Transfer learning	CNN with pre-trained models	Combining TL with CNN for recycling tasks	Larger and more diverse datasets for generalization

Year	Author(s)	Title	Journal/Conf.	Research Design	Conceptual / Theoretical Framework	Proposed Framework	Major Theme	Future Ideas
2022	Patel et al.	A multi-label waste detection model based on transfer learning	ScienceDirect	Multi-label classification using TL	Transfer learning	Multi-label CNN model	Detecting multiple waste items per image	Extend to real-time detection, broader categories
2022	Roy et al.	A Vision-Based Approach for Solid Waste Materials Feature Extraction Using Deep Learning Techniques	SpringerLink	TrashNet + custom dataset; VGG, ResNet, AlexNet	CNN architectures comparison	VGG16, VGG19, ResNet50, AlexNet comparison	Feature extraction methods	Explore ensemble across architectures
2022	Chen et al.	An automated solid waste detection using the optimized YOLO model for riverine management	Frontiers in Public Health	Object detection (YOLO optimized)	YOLO object detection framework	Optimized YOLO	Waste detection in river environments	Extend to other outdoor/urban waste monitoring systems

Year	Author(s)	Title	Journal/C onf.	Research Design	Conceptual / Theoretical Framework	Proposed Framework	Major Theme	Future Ideas
2023	Wang et al.	Application of MobileNetV2 to waste classification	PubMed	Lightweight CNN (MobileNetV2)	MobileNetV2 framework	MobileNetV2-based classifier	Domestic waste classification (4 categories)	Improve accuracy, integrate with mobile devices
2024	Li et al.	Intelligent waste classification approach based on improved multi-layered convolutional neural network	SpringerLink	Improved DCNN compared with VGG, MobileNet	DCNN, VGG, MobileNet comparisons	Improved multi-layered CNN	Organic vs recyclable waste separation	Extend to hybrid models with higher efficiency
2024	Kumar et al.	Enhanced convolutional neural network methodology for solid waste classification utilizing data augmentation techniques	ScienceDirect	Augmentation, hyperparameter tuning	CNNs with augmentation	Enhanced CNN	Performance improvement via augmentation	More novel augmentation + real-time applications
2025	Alqahtani et al.	An automated waste classification system using deep learning techniques: Toward efficient waste recycling and environmental sustainability	ScienceDirect	Dataset of 35,264 images; CNN + Ensemble ELM	Deep learning, explainable AI	Three-stage classification (biodegradable → subcategories → fine classes)	Automated classification for large datasets	Explore explainable AI for better transparency, scalability to real-world use

- In many waste management systems, segregation is performed manually by sanitation workers after waste collection. This process is time-consuming, labor-intensive, and often leads to human errors, resulting in improper waste segregation.
- Some automated approaches use basic image processing or traditional machine learning techniques that rely on handcrafted features. These methods generally have lower accuracy and struggle to classify complex or mixed waste images effectively.
- A few advanced waste sorting systems use specialized hardware such as sensors, robotic arms, or industrial sorting machines. However, these systems are expensive, require complex infrastructure, and are difficult to deploy widely in public waste management environments.

Problem Statement

- Improper waste segregation leads to environmental pollution, inefficient recycling, and increased health risks. Current systems rely heavily on manual effort or basic automation techniques that are not accurate enough.
- Therefore, there is a need for an **intelligent automated system that can classify waste images accurately and provide correct disposal guidance.**

Objectives

- To develop an **AI-based image classification system** for waste segregation.
- To classify waste into **biodegradable, recyclable, and hazardous categories.**
- To provide **confidence scores and disposal suggestions** for better waste management.

1. User Interface Module

Allows the user to upload waste images and displays classification results.

2. Image Input Module

Handles image acquisition and validates the image format.

3. Image Preprocessing Module

Resizes images to 224×224 pixels and performs normalization.

4. CNN-Based Classification Module

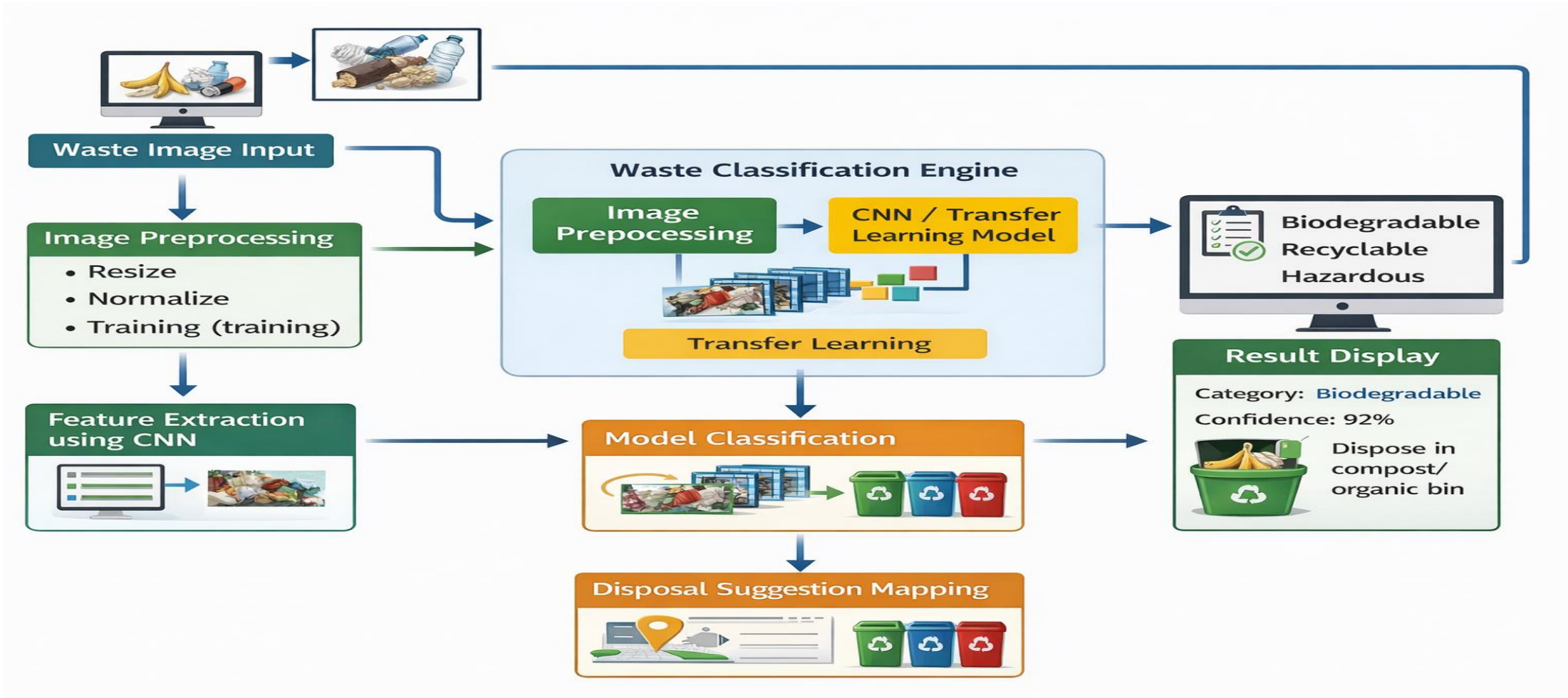
Uses EfficientNet transfer learning to classify waste images.

5. Confidence Score Module

Calculates the probability score for each prediction.

6. Disposal Suggestion Module

Maps predicted waste category to disposal instructions.



Algorithm Explanation (Training Phase)

1. Dataset Preparation

Waste images are collected and organized into three classes: **Biodegradable, Hazardous, and Recyclable**. Invalid or corrupted images are removed to maintain dataset quality.

2. Data Splitting

The dataset is divided into **training, validation, and test sets** using stratified sampling to maintain balanced class distribution.

3. Image Preprocessing and Augmentation

Images are resized to **224 × 224 pixels** and processed using **EfficientNet preprocessing**. Data augmentation techniques such as flipping, brightness adjustment, and contrast variation are applied to improve model generalization.

4. Feature Extraction using CNN (EfficientNetB0)

A **pretrained EfficientNetB0 Convolutional Neural Network** is used as the base model. CNN learns hierarchical image features through convolution operations:

$$S(i, j) = (I * K)(i, j) = \sum_m \sum_n I(i + m, j + n) K(m, n)$$

Where

I = input image,

K = convolution filter,

$S(i, j)$ = generated feature map.

5. Feature Processing and Classification Layer

Extracted features are passed through **Global Average Pooling, Dense, and Dropout layers**. A **Softmax layer** computes probabilities for the three waste classes.

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}}$$

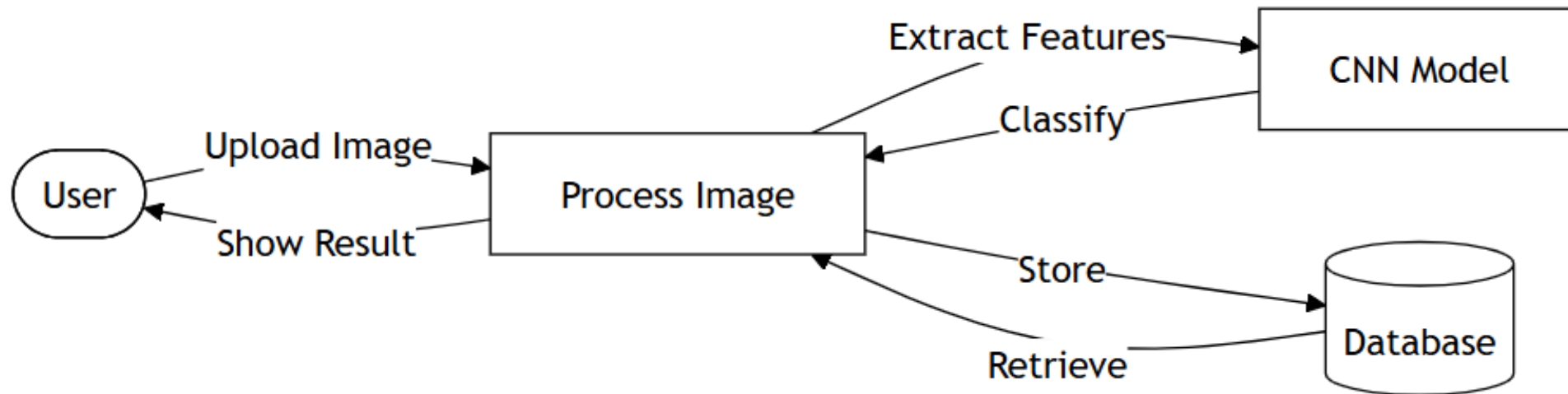
- The class with the highest probability is selected as the predicted category.

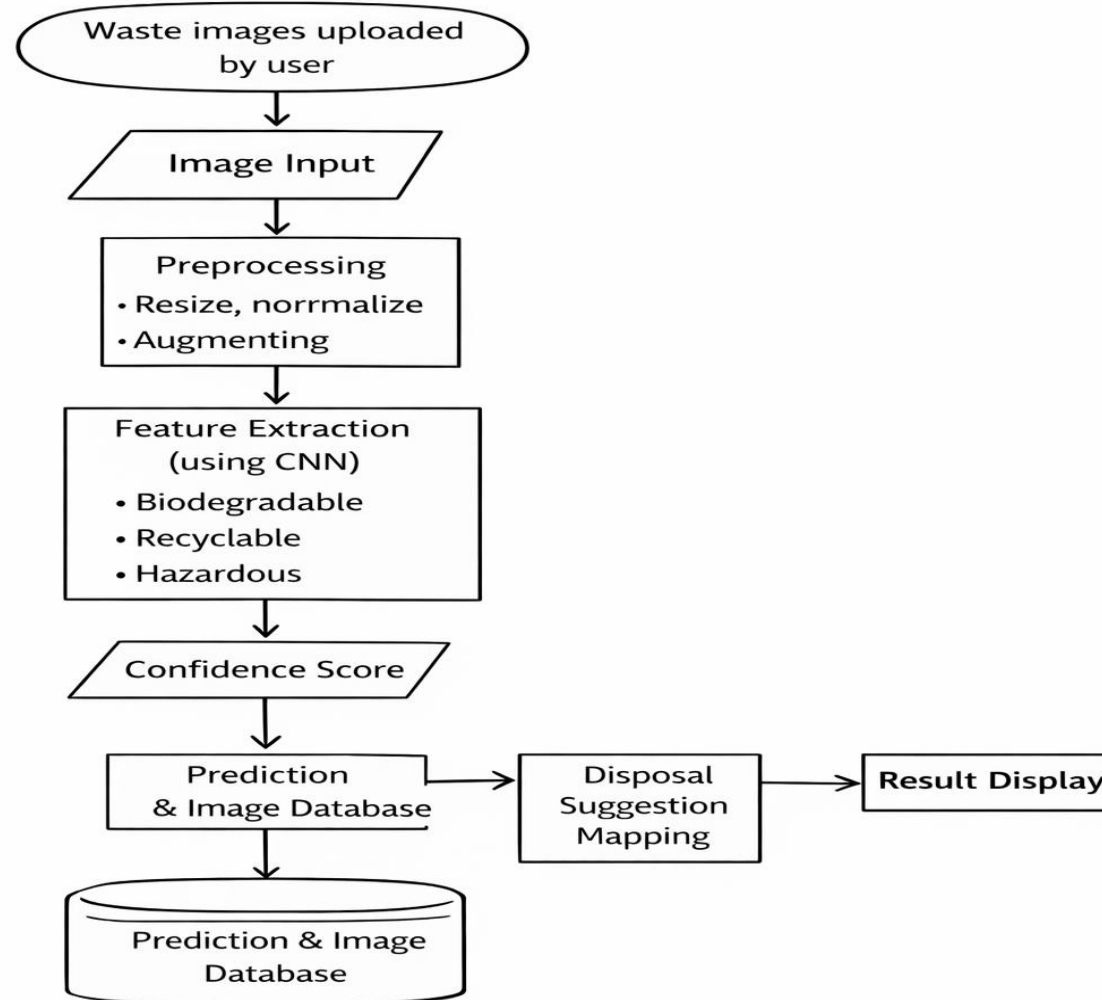
6. Model Training and Fine-Tuning

The model is trained using the **Adam optimizer** and **sparse categorical cross-entropy loss**. During fine-tuning, deeper layers of EfficientNet are unfrozen and trained with a smaller learning rate to improve classification performance.

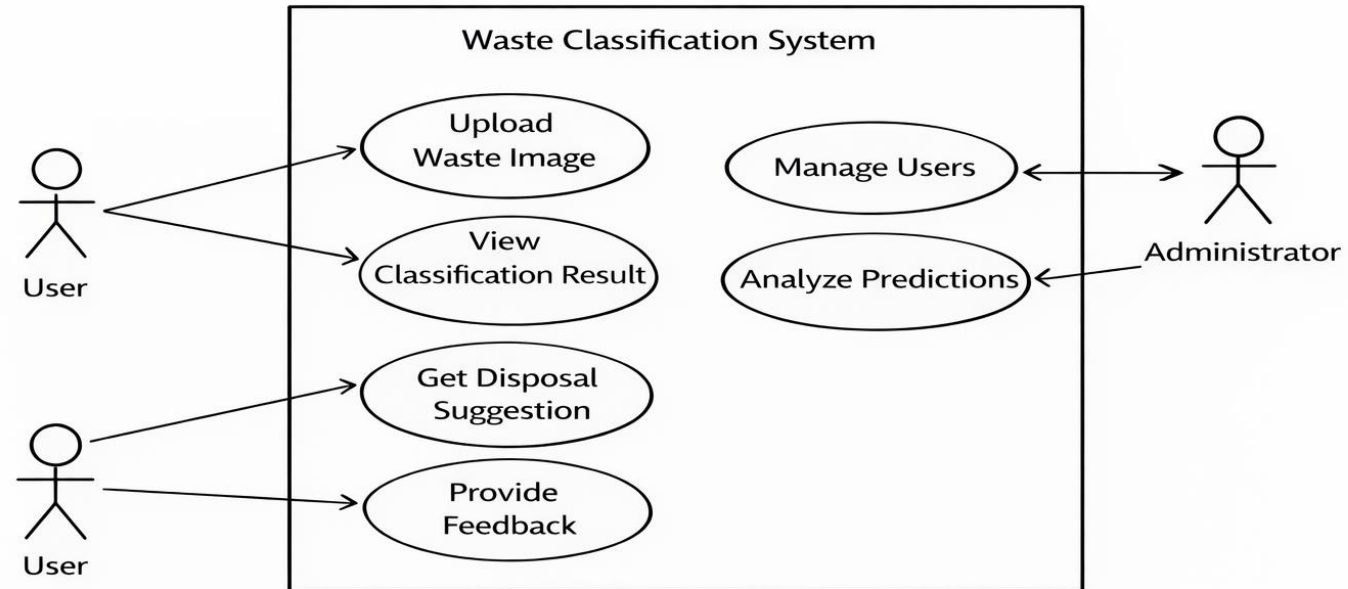
7. Prediction and Result Generation

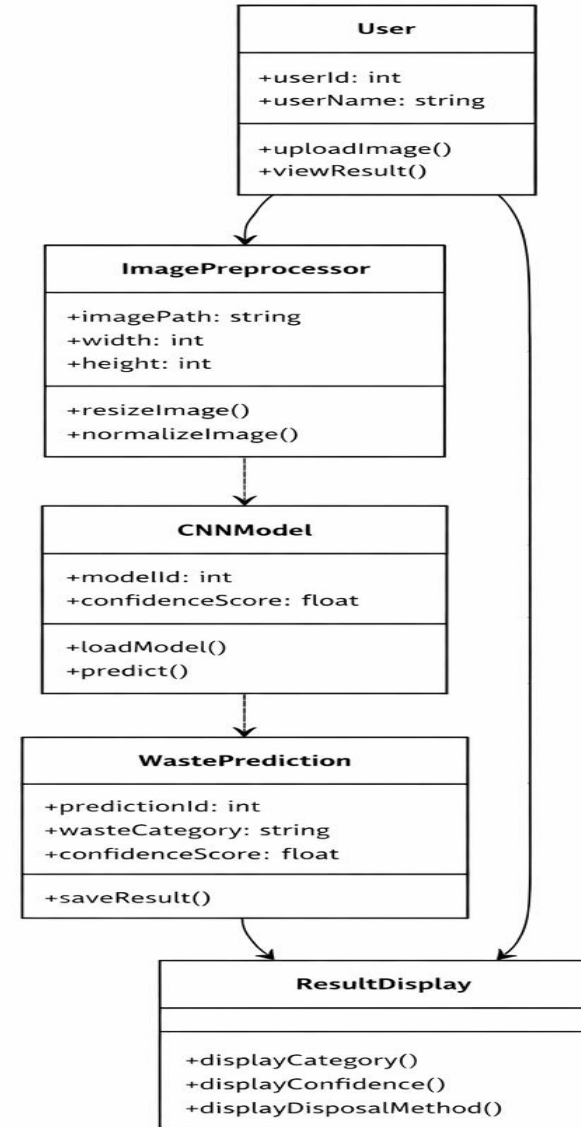
When a user uploads an image, it is preprocessed and passed to the trained model. The system outputs the **predicted waste category, confidence score, and recommended disposal method**.

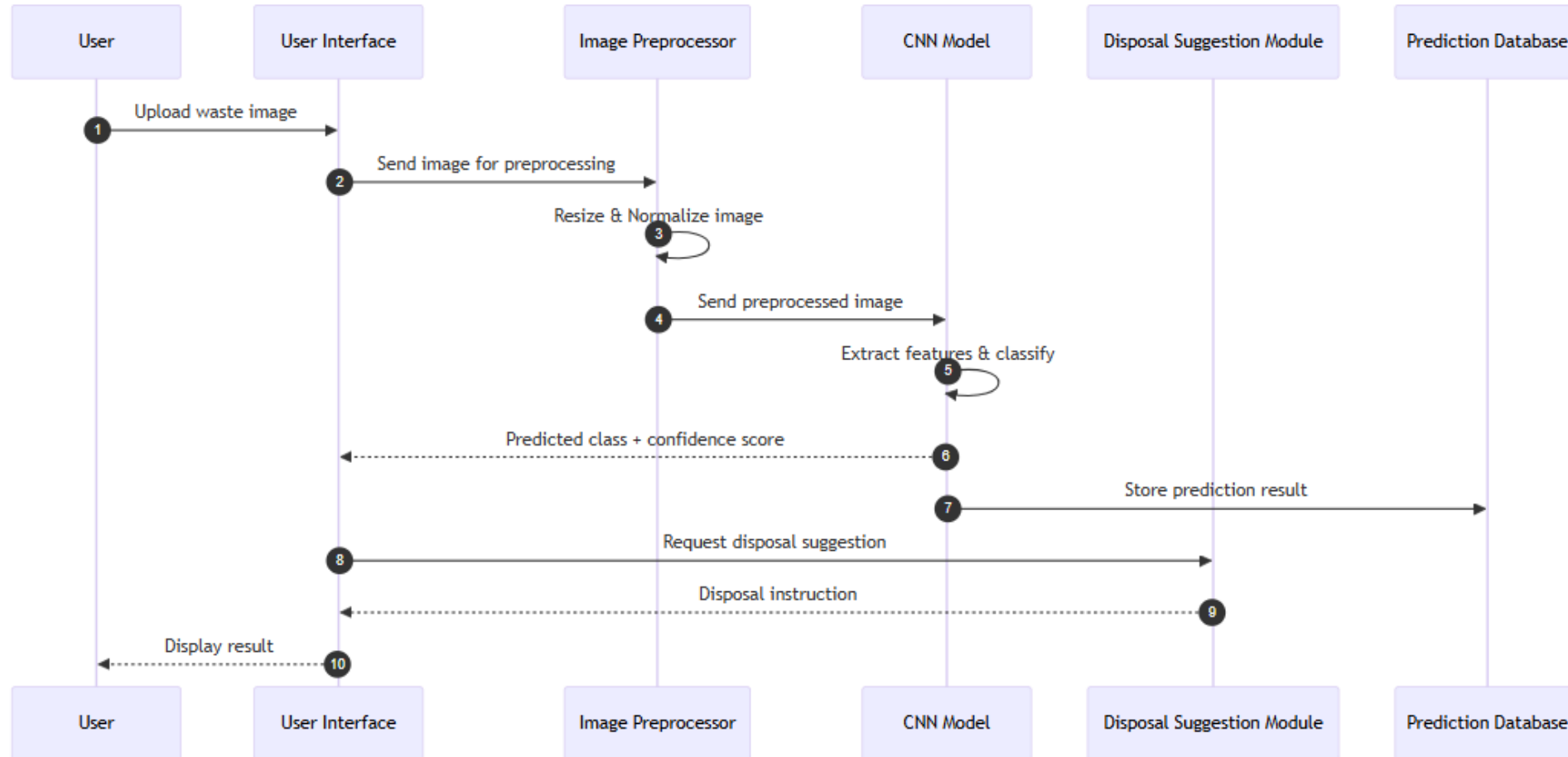


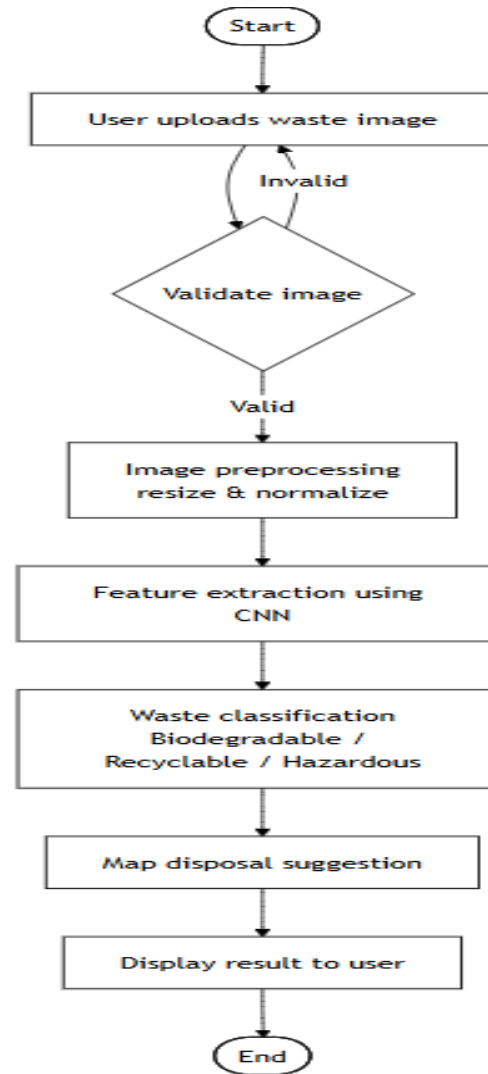


Waste Classification System - Use Case Diagram









Model Training :

The model was trained using approximately 2800 waste images divided into three classes.

```
import os

for folder in os.listdir("/content/drive/MyDrive/dataset"):
    print(folder, "->", len(os.listdir(f"/content/drive/MyDrive/dataset/{folder}")))
```

```
Recyclable -> 630
Hazardous -> 1537
Biodegradable -> 647
```

Since the dataset was imbalanced, class weights were applied during training so that minority classes receive higher importance and the model does not become biased toward the majority class.

Model Performance :

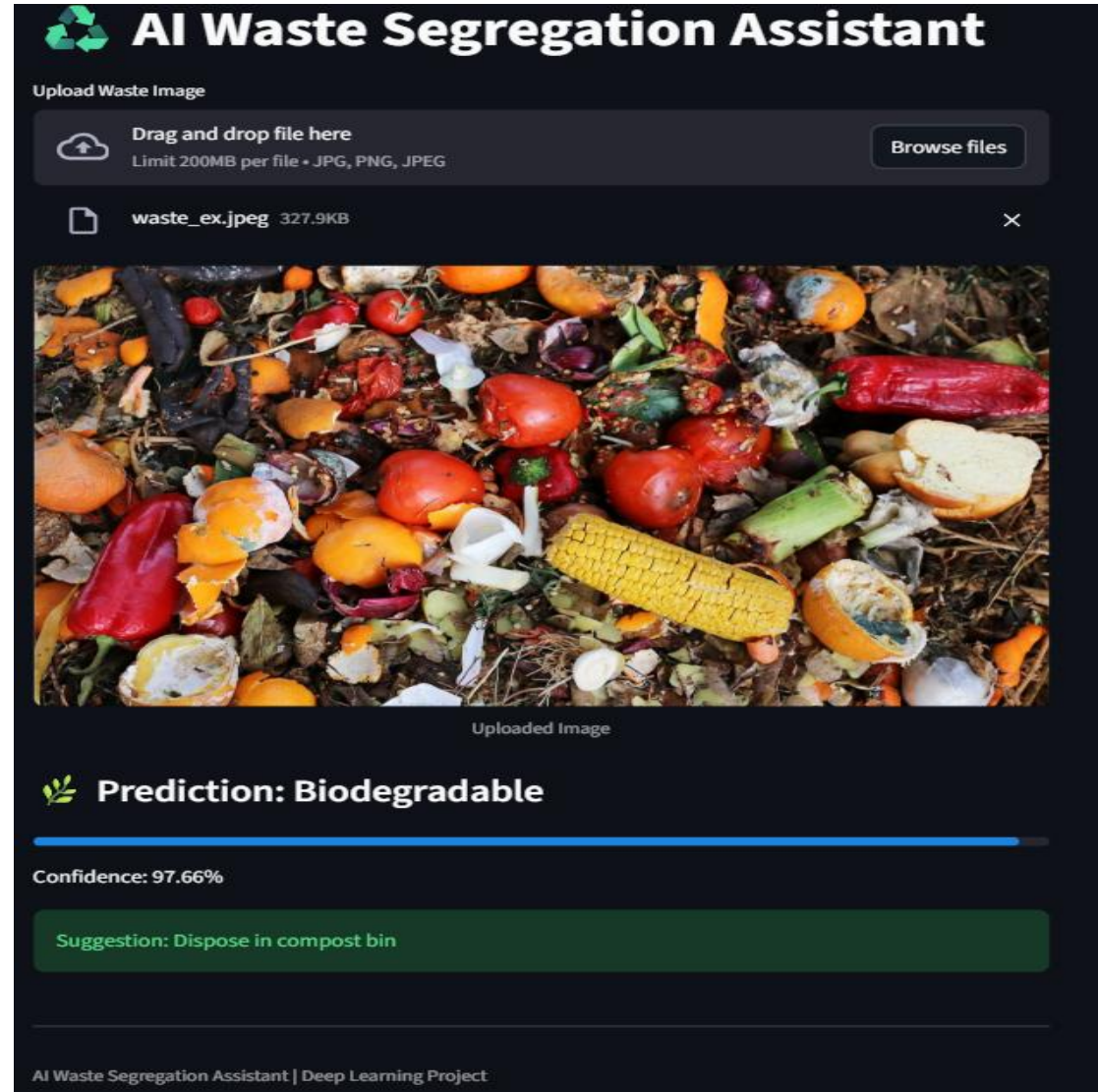
Classification Report:

	precision	recall	f1-score	support
Biodegradable	0.82	0.86	0.84	97
Hazardous	0.89	0.81	0.85	231
Recyclable	0.67	0.80	0.73	95
accuracy			0.82	423
macro avg	0.79	0.82	0.80	423
weighted avg	0.83	0.82	0.82	423

- Accuracy measures overall correctness of predictions.
- Precision indicates correctness of predicted classes.
- Recall measures how many actual waste items were correctly identified.

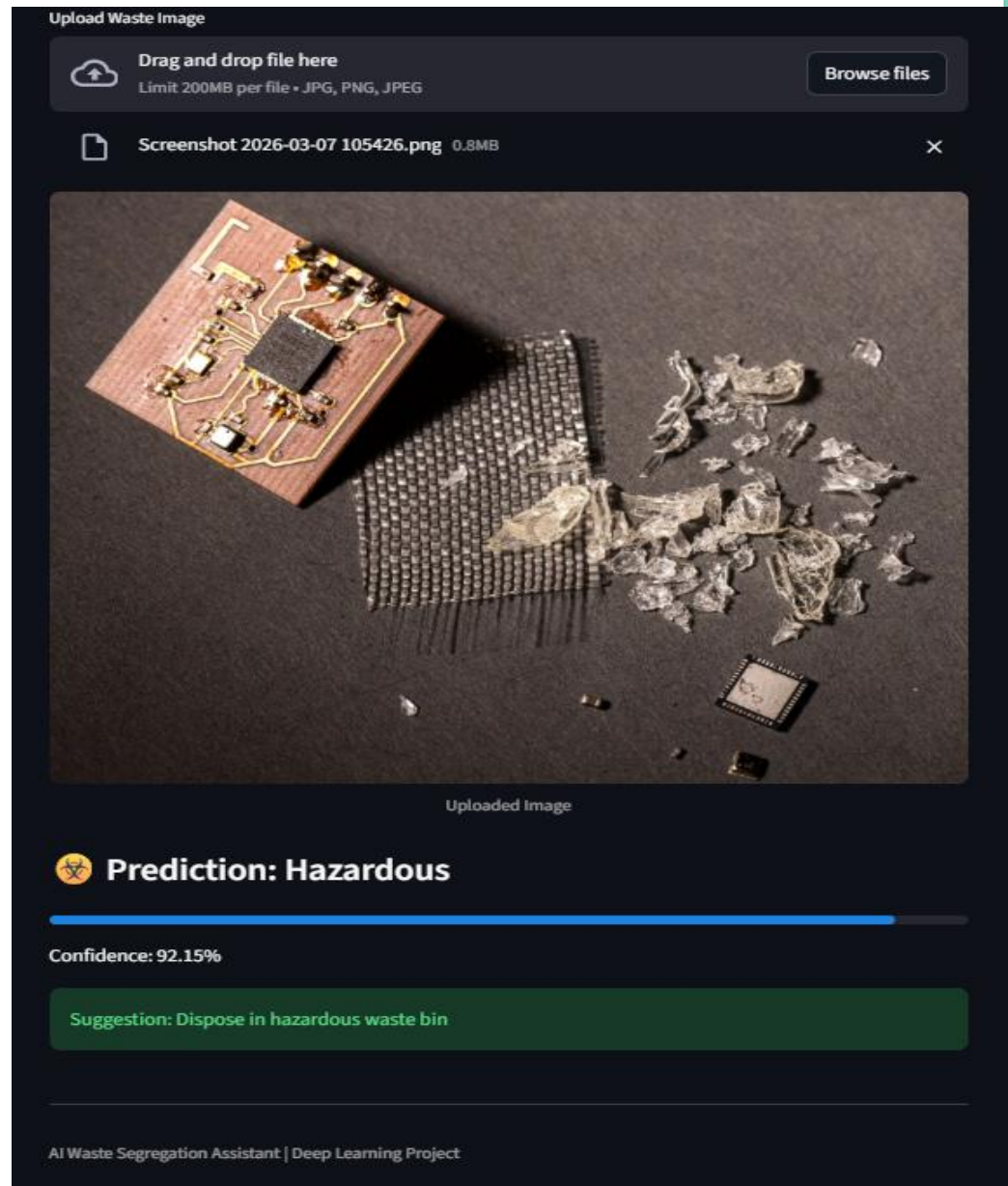
System Interface:

The user uploads a waste image, and the system displays prediction, confidence score, and disposal suggestion



Example Prediction:

For example, electronic components were classified as hazardous waste with a confidence score of around 92.15%, and the system suggested disposing them in a hazardous waste bin.



Conclusion :

- The project successfully developed an **AI-based waste classification system using CNN and transfer learning.**
- The system can accurately classify waste images and provide correct disposal suggestions.

Future Enhancement :

- Integration with **smart waste bins for automatic waste segregation.**
- Development of **mobile applications for real-time waste classification.**
- **Extension to Waste Detection Systems:** The proposed classification model can be extended to object detection models to identify and localize multiple waste items within a single image.

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Thank You